



Digital Communication IC Design

Chap 4: Source Coding, Channel Coding, and Decoding

Chih-Feng Wu (吳志峰), PhD.

*Department of Electronics Engineering, Chang Gung University,
Taoyuan, Taiwan (ROC)*

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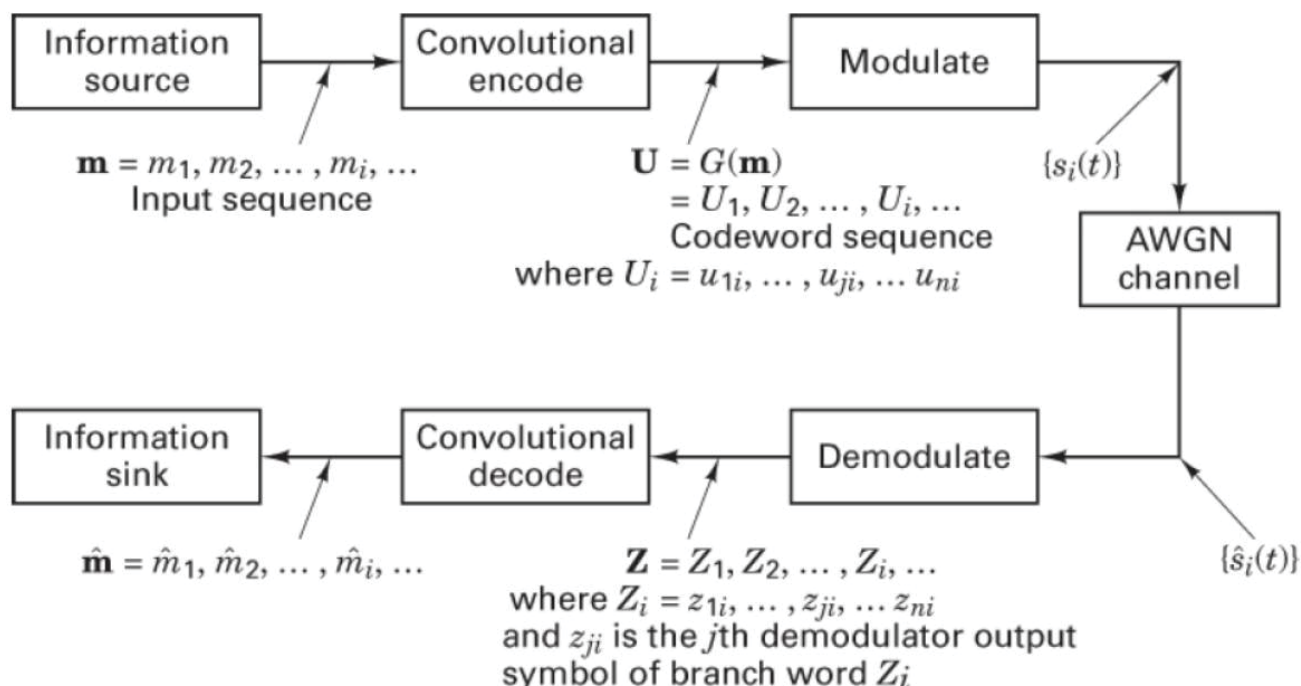


Outline

- ☐ Convolutional Code
- ☐ Viterbi Algorithm
- ☐ Trellis Decoding

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Channel Coding and Decoding



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Channel Coding

- ❑ Two major channel codes
 - Block
 - Convolutional
- ❑ Linear block code
 - Two integers, n and k , and a generator matrix or polynomial
 - k : the number of data bits form an input to block encoder
 - n : the total number of bits for the outputs of the encoder
 - k/n : code rate of the linear block code
 - $(n - k)$: the number of the redundant bits
- ❑ Convolution code
 - Three integers, n , k and K
 - K : the constraint length that is the number of the shift registers
 - k/n : code rate (the information of coded bit)

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Convolutional Encoding

- ❑ Input message sequence m_i ($\mathbf{m} = m_1, m_2, \dots, m_i$)
 - Binary digit (bit): 1 or 0
- ❑ Codeword sequence U_i ($\mathbf{U} = G(\mathbf{m}) = U_1, U_2, \dots, U_i$)
 - U_i : Made up of coded bits that are not independent
- ❑ Demodulator output Z_i ($\mathbf{Z} = Z_1, Z_2, \dots, Z_i$)
- ❑ Decoded sequence $\hat{m}_i$ ($\hat{\mathbf{m}} = \hat{m}_1, \hat{m}_2, \dots, \hat{m}_i$)

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Convolutional Encoder

- ❑ General convolution encoder (n, k, K)

- Shift register

- Modulo-2 adder (XOR)

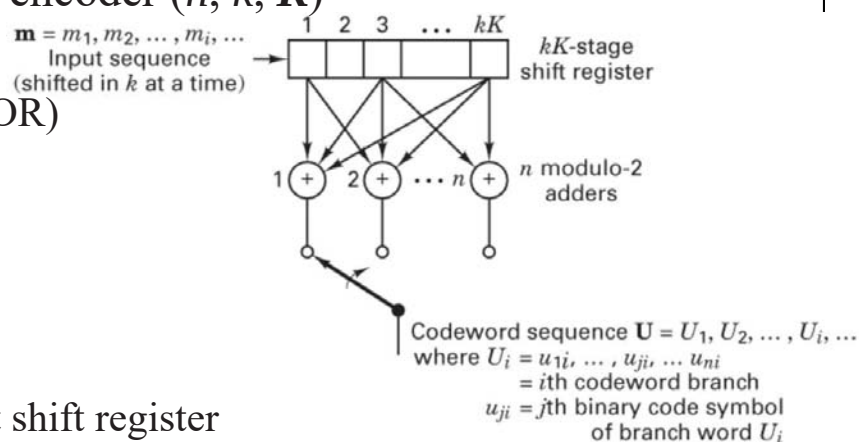
- $0 \oplus 0 = 0$;
- $0 \oplus 1 = 1$;
- $1 \oplus 0 = 1$;
- $1 \oplus 1 = 0$;

- ❑ K constraint length

- The number of k -bit shift register

- ❑ Code rate: k/n

- Shift k -bit into shift register
- Generate n coded bits
- $k < n$



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Convolutional Encoder (cont'd)

□ Encoding function $G(\mathbf{m})$

■ $\mathbf{U} = G(\mathbf{m})$

■ Use state diagram (or tree diagram, trellis diagram) to represent a convolutional encoder

■ Connection vector or polynomial

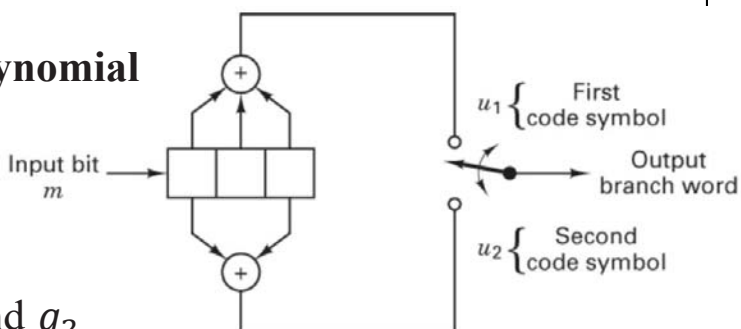
□ $\frac{1}{2}$ convolutional encoder

■ code rate: $\frac{1}{2}$ ($k=1, n=2$)

■ Connection vectors: g_1 and g_2

• $g_1 = [1 \ 1 \ 1] = 7_8$ and $g_2 = [1 \ 0 \ 1] = 5_8$

■ Constraint length determined by the dimension of connection vectors



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Convolutional Encoder (cont'd)

Input sequence:

1 $\rightarrow t_1$

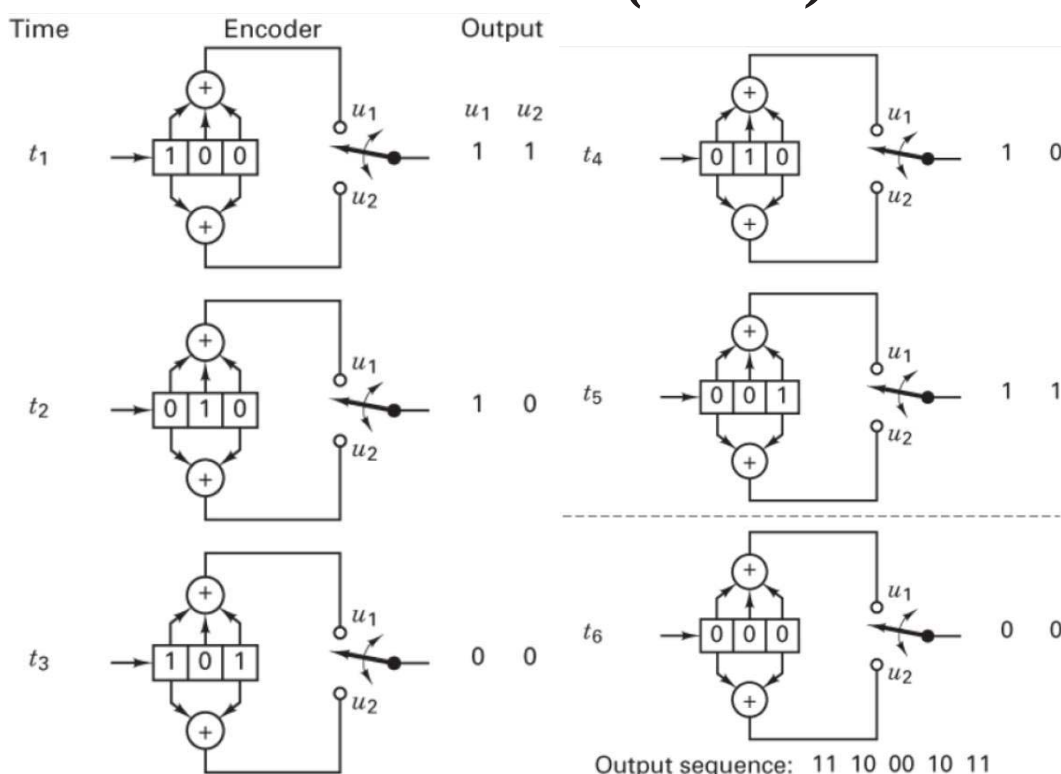
0

1

0

0

0 $\rightarrow t_6$



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Polynomial Representation and State Diagram

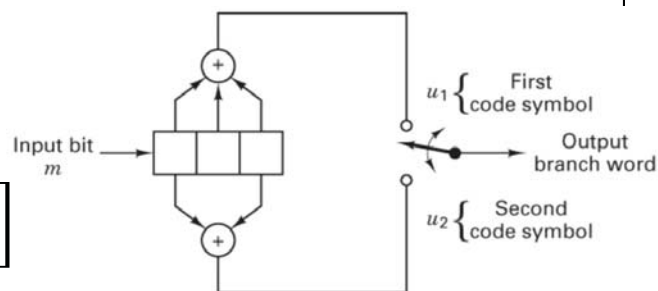
□ Convolutional encoder (2, 1, 3)

- n generator polynomials, K shift registers and modulo-2 adder
- Each polynomial with the degree of $K-1$ or less
- Generator polynomials

$$g_1(X) = 1 + X + X^2$$

$$g_2(X) = 1 + X^2$$

$$\mathbf{G} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7_o \\ 5_o \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$



□ State Diagram

- The number of state defined as the contents of the rightmost $K-1$ stages
 - 2-stage $\rightarrow$ 4 stage-diagram

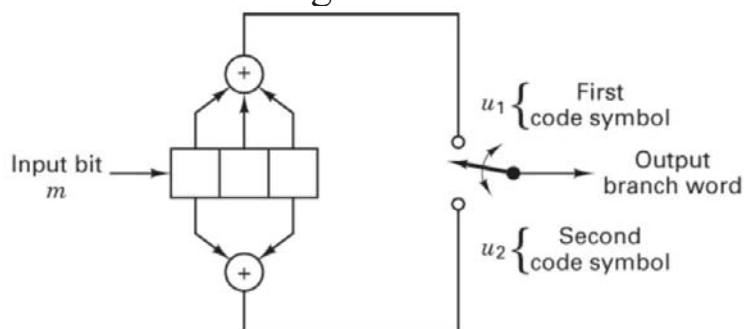
State Diagram (cont'd)

$$\mathbf{U} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = G(m)$$

State Diagram (cont'd)

Example: (2, 1, 3) convolution encoder

- A convolutional encoder with code rate $\frac{1}{2}$ ($k=1, n=2, K=3$)
- Generator polynomials: $g_1 = [1 \ 1 \ 1] = 7_8$ and $g_2 = [1 \ 0 \ 1] = 5_8$
- The message sequence $\mathbf{m} = 1, 1, 0, 1, 1$, followed by 2 zeros to flush the shift registers
- Assume the initial all zeros of the shift registers



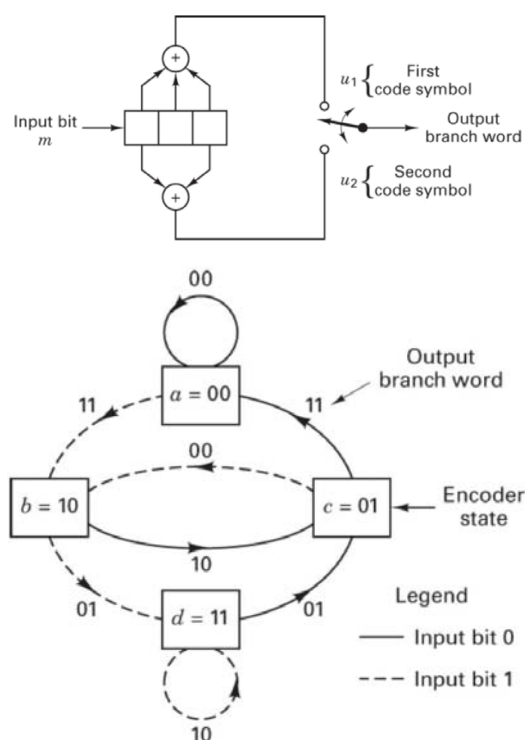
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State Diagram (cont'd)

Input bit: $m_i = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

Input bit m_i	Register contents	State at time t_i	State at time t_{i+1}	Branch word at time t_i	
				u_1	u_2
—	000	00	00	—	—
1	100	00	10	1	1
1	110	10	11	0	1
0	011	11	01	0	1
1	101	01	10	0	0
1	110	10	11	0	1
0	011	11	01	0	1
0	001	01	00	1	1



Output sequence: $U = 11 \ 01 \ 01 \ 00 \ 01 \ 01 \ 11$

State Diagram (cont'd)

□ Assume the shift registers without all zeros initial contents

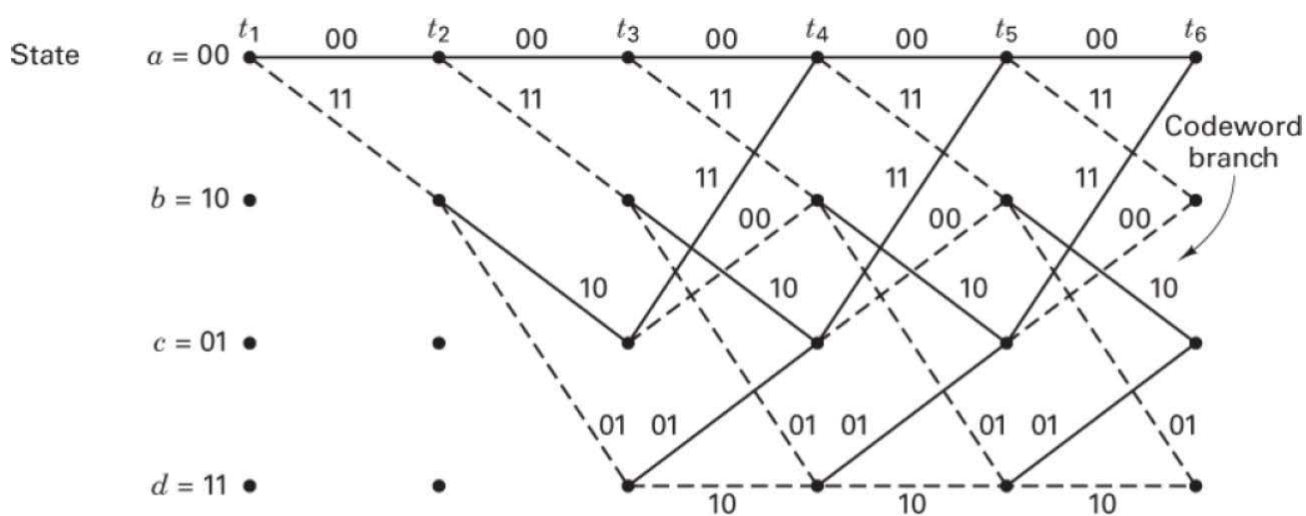
■ Input sequence: $\mathbf{m} = 1, 1, 0, 1, 1, 0, 0$

Input bit m_i	Register contents	State at time t_i	State at time t_{i+1}	Branch word at time t_i	
				$u1$	$u2$
—	11 ×	1 ×	11	—	
1	111	11	11	1	0
1	111	11	11	1	0
0	011	11	01	0	1
1	101	01	10	0	0
1	110	10	11	0	1
0	011	11	01	0	1
0	001	01	00	1	1

Output sequence: $\mathbf{U} = 10\ 10\ 01\ 00\ 01\ 01\ 11$
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Encoder Trellis Diagram



Input sequence: $\mathbf{m} = 1, 1, 0, 1, 1, 0, 0$
Output sequence: $\mathbf{U} = 10\ 10\ 01\ 00\ 01\ 01\ 11$



Viterbi Convolutional Decoding Algorithm

- ❑ Discovered and analyzed by Viterbi in 1967
 - Perform maximum likelihood decoding
 - Calculate a *measure of similarity*, or *distance* between the *received signal* at time t_i , and all the *trellis paths entering* each state at time t_i
- ❑ Decoding
 - Hard-decision
 - Find the path with the **minimum Hamming distance** from the received signal
 - $d(c, y) = \sum_{i=1}^m d(c_i, y_i)$, c_i : Trellis codeword; y_i : the quantized received signal;
 - Soft-decision
 - Find the path with the **minimum Euclidean distance** from the received signal

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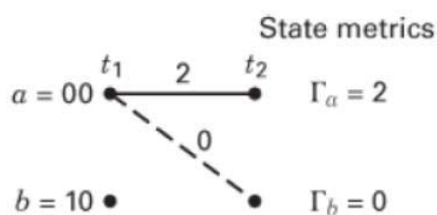
Viterbi Convolutional Decoding Algorithm(cont'd)

- ❑ Hamming distance
 - Considering two strings with equal length, the Hamming distance is defined as the number of positions at which the corresponding **symbols** are different.
 - For example:
 - 0000 and 1111 → Hamming distance = 4
 - 1011101 and 1001001 → Hamming distance = 2
 - “**t**oned” and “**r**oses” → Hamming distance = 3

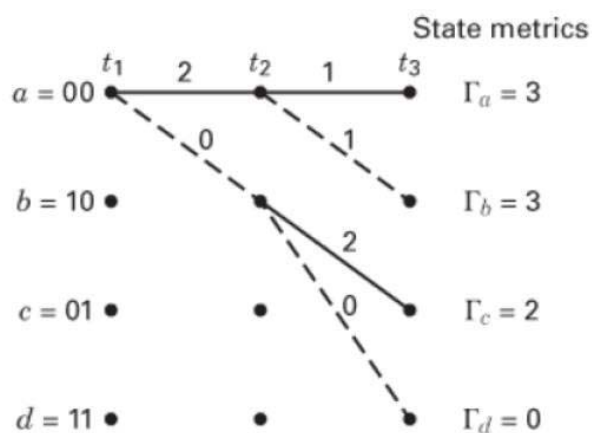
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Viterbi Decoding Process



(a) $Z = 11$



(b) $Z = 01$

Input sequence: $m = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

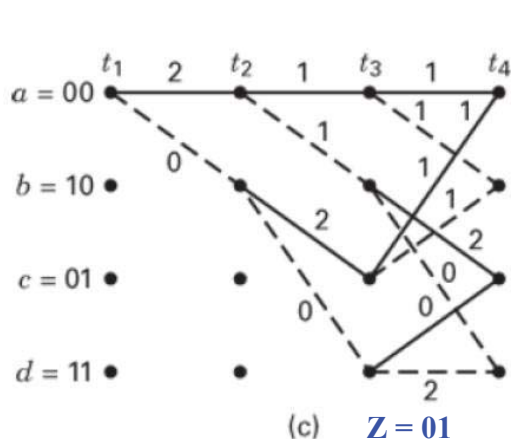
Output sequence: $U = 11 \ 01 \ 01 \ 00 \ 01 \ \dots$

Received sequence: $Z = 11 \ 01 \ 01 \ 10 \ 01 \ \dots$

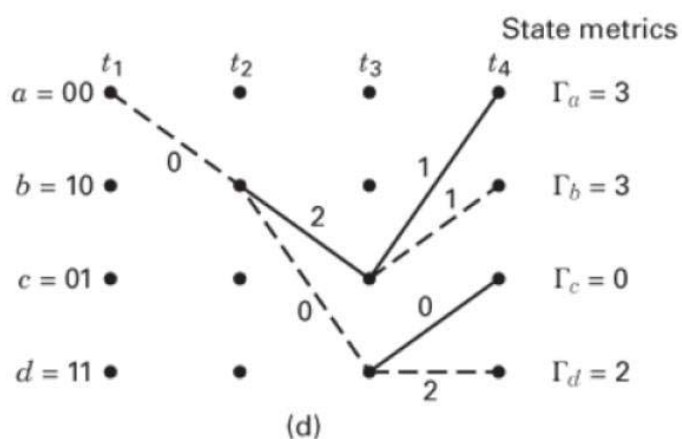
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Viterbi Decoding Process (cont'd)



(c) $Z = 01$



(d)

Input sequence: $m = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

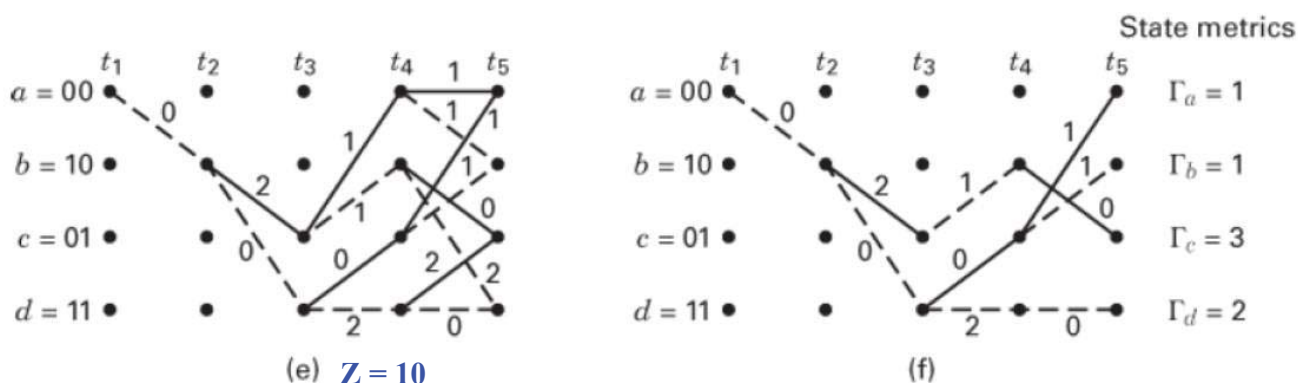
Output sequence: $U = 11 \ 01 \ 01 \ 00 \ 01 \ \dots$

Received sequence: $Z = 11 \ 01 \ 01 \ 10 \ 01 \ \dots$

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Viterbi Decoding Process (cont'd)



Input sequence: $m = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

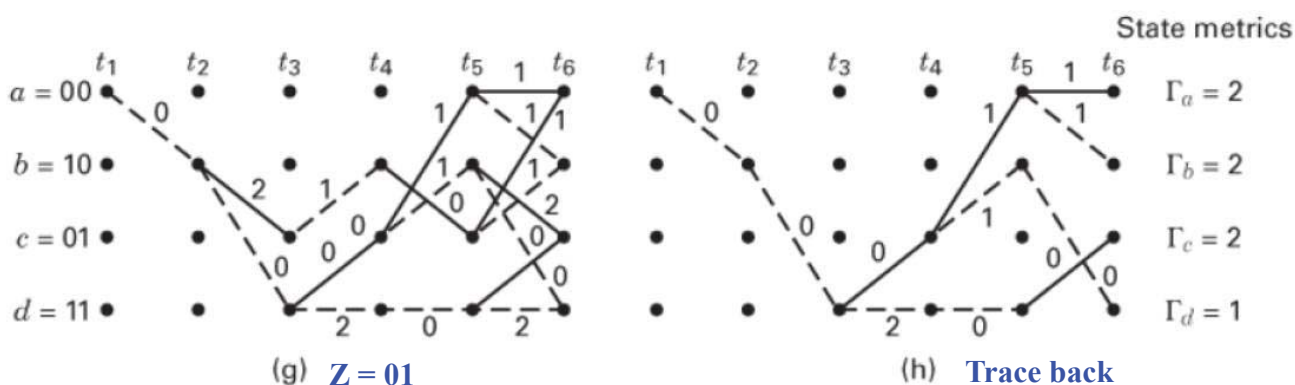
Output sequence: $U = 11 \ 01 \ 01 \ 00 \ 01 \ \dots$

Received sequence: $Z = 11 \ 01 \ 01 \ 10 \ 01 \ \dots$

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Viterbi Decoding Process (cont'd)



Input sequence: $m = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

Output sequence: $U = 11 \ 01 \ 01 \ 00 \ 01 \ \dots$

Received sequence: $Z = 11 \ 01 \ 01 \ 10 \ 01 \ \dots$

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